

Asset Inspection Data Quality and Anomaly Detection



[UC56] - Awesense Platform Use Case

The Utility Problem

Field inspection programs are one of the primary mechanisms utilities use to verify the physical condition of their distribution assets. Inspectors visit substations, pole-mounted transformers, switching equipment, and underground infrastructure on a regular cycle, recording observations about insulation type, condition ratings, oil levels, visible defects, and other physical characteristics. These inspection records are then filed — as paper forms, PDFs, photographs, or structured reports — and stored separately from the master asset data held in the GIS.

The central problem is that inspection findings are rarely cross-referenced against the master asset record in any systematic way. A GIS record may indicate that a transformer is air-insulated, while the inspector's report clearly documents an oil-insulated unit at the same location. A switchgear entry may list one voltage class, while the nameplate photograph taken during the last inspection shows a different rating. These discrepancies accumulate silently over time — eroding the reliability of the GIS as a source of truth, distorting capacity analysis, and introducing risk into engineering decisions that depend on accurate asset attributes.

Compounding the problem, inspection findings themselves are not monitored for internal consistency over time. A condition that was rated as a major defect in one inspection cycle may reappear as minor or absent in the next — a reversal that could indicate a data entry error, a substituted asset that was never recorded in the GIS, or a genuine remediation that was never formally closed out in the work management system. Without automated monitoring of inspection result sequences, these anomalies go undetected.

The root cause lies in the nature of inspection data itself: it is predominantly unstructured. Inspection reports exist as PDF documents, scanned forms, annotated photographs, and free-text technician notes — formats that do not lend themselves to direct comparison against the structured attribute fields of a GIS record. Reconciling the two has historically required manual review by experienced engineers, which is both expensive and difficult to perform at scale across a large asset fleet.

The Data Solutions and Methods

The Awesense Platform provides three foundational capabilities that together enable a structured approach to this problem. First, it maintains the Awesense Energy Data Model (EDM) — a structured, unified data layer that normalizes asset attribute data ingested from the utility's GIS, OMS, and other source systems into a consistent, queryable representation of the grid. Second, the EDM is fully queryable, enabling analytical workflows to systematically interrogate asset attributes across the entire fleet. Third, the platform supports the attachment of unstructured data — PDF inspection reports, field photographs, scanned condition assessment forms, and other documents — directly to the corresponding GIS asset object within the EDM.

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Each attachment is persistently linked to its asset's unique identifier, creating a co-located pair of structured GIS attributes and unstructured inspection evidence for every asset in the fleet.

This combination of structured data and co-located unstructured attachments creates the foundation for a discrepancy detection workflow. The structured side of the problem is already solved by the EDM: asset attributes such as insulation type, voltage class, manufacturer, and phase configuration are queryable and comparable across the fleet. The remaining challenge is extracting equivalent structured information from the unstructured inspection documents attached to each asset.

One promising technique for bridging this gap is the application of Large Language Models (LLMs) to the attached inspection content. Applied to a PDF inspection report, an LLM could read the document and extract key asset attributes mentioned in the text — insulation type, observed voltage class, manufacturer, condition rating, defect descriptions — returning them as structured fields. Applied to field photographs, a vision-capable model could read nameplate data, identify equipment characteristics from visual cues, and flag visible physical anomalies. The extracted attributes could then be compared field-by-field against the corresponding EDM record for that asset, with mismatches surfaced as data quality flags. Flags could be prioritized by severity — a discrepancy in insulation type or voltage class escalated immediately, minor attribute differences queued for routine review.

A complementary analytical technique is longitudinal monitoring of inspection results over time using the structured data already captured in the EDM. For each asset, the sequence of condition ratings and defect records across inspection cycles can be queried and analyzed for anomalous reversals — for example, a defect rated as major in one cycle that appears as none or minor in the next, without a corresponding closed work order in the system. Such reversals are strong indicators of data entry errors, unreported asset replacements, or remediations that were never formally documented — and can be detected entirely within the structured EDM without requiring unstructured document analysis.

Together, these approaches — grounded in the Awesense Platform's structured EDM and unstructured data attachment capability, and extended by emerging LLM techniques — offer utilities a scalable path to systematically closing the gap between field reality and master data records.

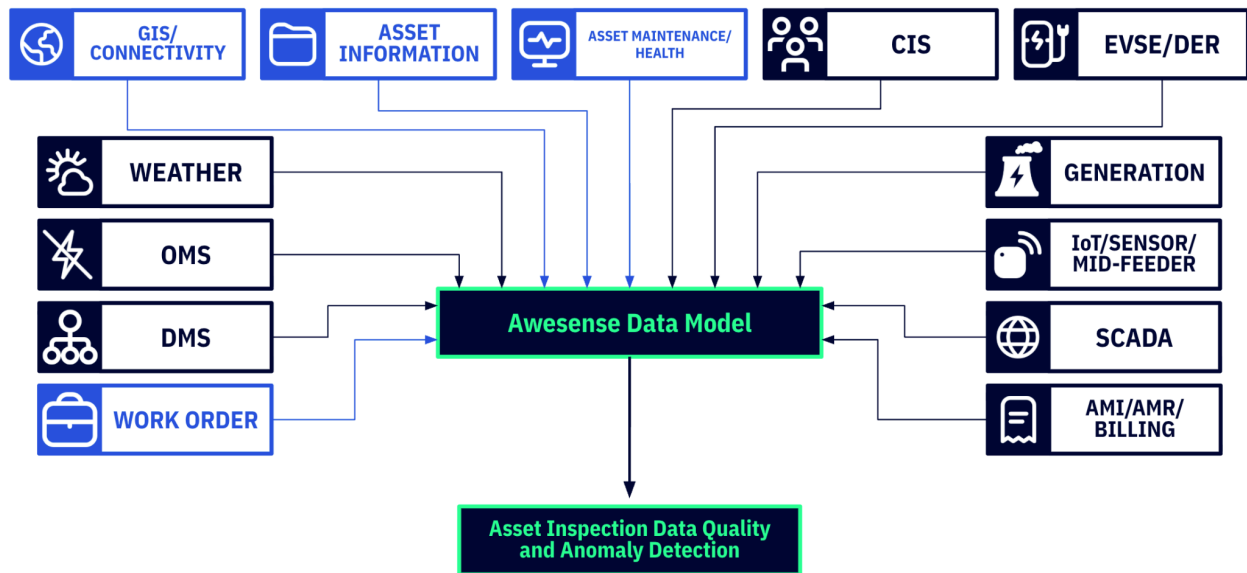
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Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.